# STUDENT PROJECTS LABORATORY - LOCKOUT / TAGOUT [LOTO] PROCEDURE

**ENERGY ISOLATION SPECIFICATION | ASSET: WLD-05 | TAG: THERM-BM-MF**

## Thermal Processing Suite: Blue M Industrial Oven & Muffle Furnace

**Authorized LOTO Personnel: Sr. Instructional Lab Supervisor (Max Wright) & Student Research Assistants (SRAs)**

Notification Channel: Max Wright - Teams / Text Message | Zone: Zone 3: Welding & Thermal Processing (WLD)

## STEP 1 - NORMAL MACHINE SHUTDOWN PROCEDURE

Blue M Oven: Lower setpoint to 167 deg F (75 deg C) and allow circulation fan to run until cooled below 167 deg F (turning off above 167 deg F seizes circulation motor bearings!), then toggle power switch OFF (Front lower center). Muffle Furnace: Press RUN key to stop heating program, toggle main power switch OFF.

## STEP 2 - HAZARDOUS ENERGY SOURCES, MAGNITUDES & ISOLATION POINTS

Source #1: Electrical (Blue M Oven) (240V AC)

Isolation Location: Wall switch box disconnect adjacent to Blue M Oven.

Lockout Procedure: Pull wall switch box lever down to OFF & apply LOTO lock; verify chamber light does not illuminate.

Source #2: Thermal (Blue M Oven) (Up to 650 deg F (343 deg C))

Isolation Location: Internal stainless steel heating chamber.

Lockout Procedure: Allow chamber to cool below 100 deg F / ambient before performing any internal maintenance.

Source #3: Electrical (Muffle Furnace) (120V AC)

Isolation Location: 120V power cord & wall receptacle.

Lockout Procedure: Unplug power cord and secure male plug inside Rotating Electrical Plug Lockout box; verify digital display is dark.

Source #4: Thermal (Muffle Furnace) (Up to 1,250 deg C (2,282 deg F))

Isolation Location: Refractory ceramic furnace chamber.

Lockout Procedure: Keep door closed until internal temp is < 200 deg C, then crack door slightly to reach ambient room temperature before servicing.

## STEP 3 - STORED / RESIDUAL ENERGY DISSIPATION

Blue M Oven: Mandatory fan-on cooldown < 167 deg F (75 deg C) then cool to < 100 deg F. Muffle Furnace: Closed-door cooldown < 200 deg C to prevent refractory thermal shock, then cool to ambient.

## STEP 4 - ZERO-ENERGY VERIFICATION ('TRY' STEP)

Verify Blue M chamber light and Muffle Furnace controller stay completely dark and chamber thermometer reads ambient (<100 deg F).